

Designing and Handling Guidline for CX70M-24P1

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1. Revision History

Revision History	Date	Handled by	Remarks
1.0	June. 27, 2018	JH.BOO	Draft version
2.0	April. 15, 2022	SK.JANG	Revised

2. Introduction

2.1 Purpose

The guidelines are intended to provide information on product features and how to handle them.

Guidelines are intended to provide general information and do not limit your design or guarantee results in all situations.

2.2 Scope

Guidelines describe basic design information, recommended device dimensions, and regulatory requirements.

These guidelines will be revised from time to time to reflect changes in technology and production capacity.

2.3 Reference Specification

- Universal Serial Bus Type-C Cable and Connector Specification
Revision 2.1 May 2021

3. Product Information

3.1 Product Feature

- * USB3.0(5Gbps), high speed transmission
- * Minimized length design for space saving
 - Length : 8.35mm
- * Reversible plug orientation ensures easy insertion
- * Hybrid mounting design provides easy inspection and rework
- * Compliant to USB specification
 - USB Type-C compliant interface connector.

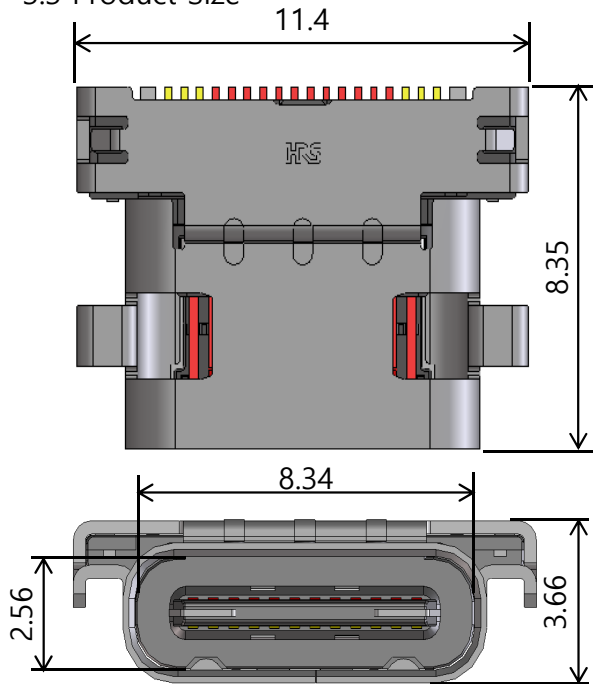
3.2 Specification

No. of Contacts	24
P.C.B Mounting type	Mid-mount
Soldering type	Right angle Hybrid (SMT+Dip)
Current rating	DC 5A max. for Vbus & GND DC 1.25A for Vconn DC 0.25A for the other pins
Voltage rating	48V AC/DC
Operating Temperature	-40°C ~ +105°C (Including Temp.rise), 95% RH MAX
Storage Condition	-10°C ~ +60°C (With Packing), 15% ~ 70% RH
Contact Resistance	40mΩ Max. (Initial)
Withstanding Voltage	100V AC for 1 minute
Insulation Resistance	100MΩ Min. (500V DC)
Mating Cycles	10,000 times
Insertion/ Extraction Force	Insertion: 5~20N, Extraction: 8~20N

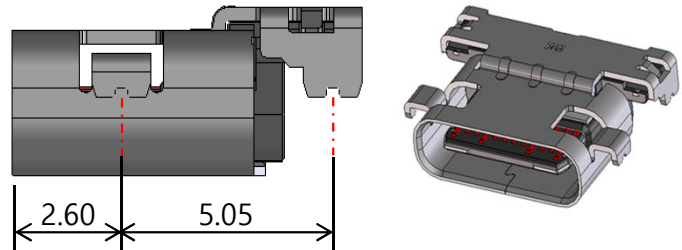
Note

- * Storage conditions apply to original packaging only, void if opened. Warranty period is 12month max. in the storage conditions above and calculated by manufacture date code.

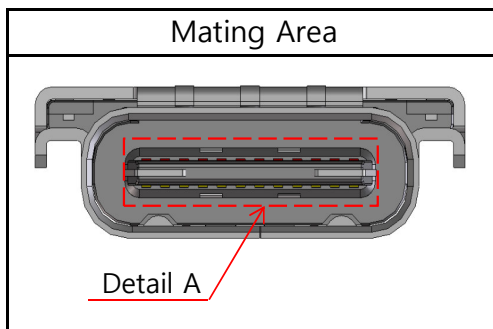
3.3 Product Size



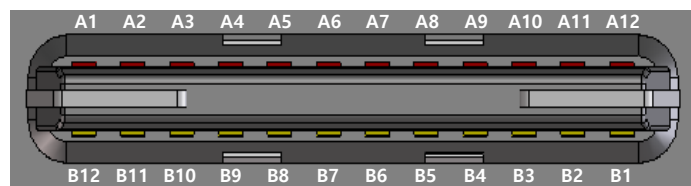
Width	11.4
Length	8.35
Height	3.66



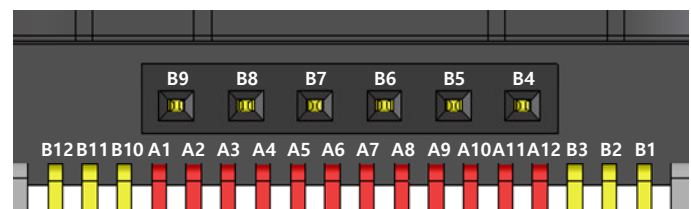
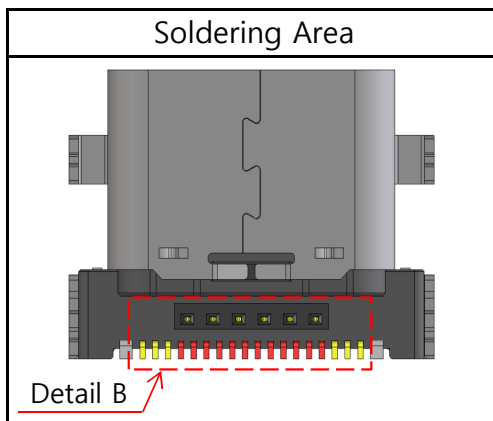
3.4 Pin Assignment



A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
GND	TX1+	TX1-	Vbus	CC1	D+	D-	SBU1	Vbus	RX2-	RX2+	GND
B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1
GND	RX1+	RX1-	Vbus	SBU2	D-	D+	CC2	Vbus	TX2-	TX2+	GND

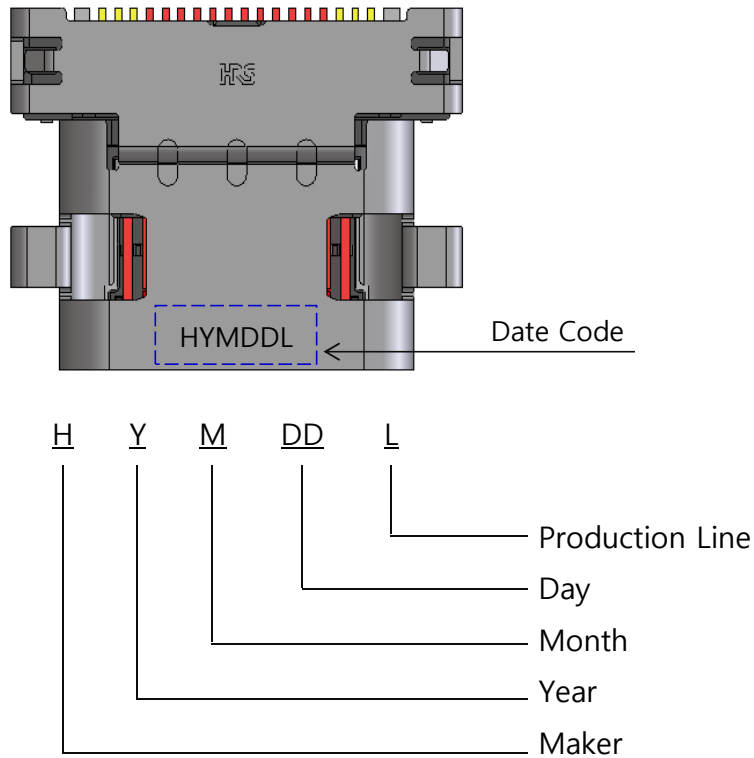


Detail A



Detail B

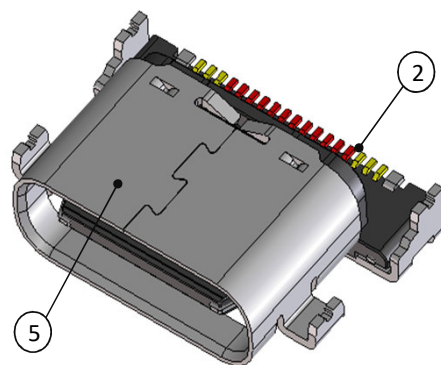
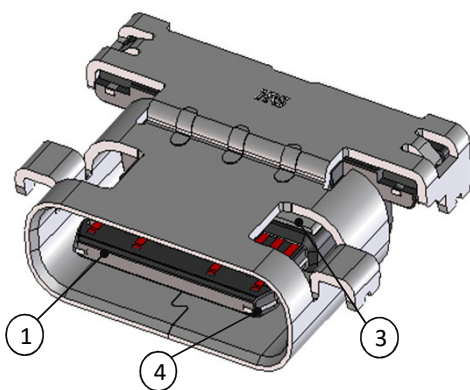
3.5 Manufacturing Date Code System



Maker		Year		Month		Day		Production Line	
Ex.	Mark	Ex.	Mark	Ex.	Mark	Ex.	Mark	Ex.	Mark
HiroseKorea	H	2018	8	Jan.	1	1st	01	SAMPLE	S
		2019	9	Feb.	2	2nd	02	Manual #1	1
		2020	0	Mar.	3	3rd	03	Manual #2	2
		2021	1	Apr.	4	4th	04	Manual #3	3
		2022	2	May	5	5th	05	Manual #4	4
		2023	3	Jun.	6	6th	06	...	...
		2024	4	Jul.	7	7th	07	Auto #1	A
		2025	5	Aug.	8	8th	08	Auto #2	B
		2026	6	Sep.	9	9th	09	Auto #3	C
		2027	7	Oct.	A	10th	10	Auto #4	D
		...	...	Nov.	B	11th	11	Auto #5	E
		...	...	Dec.	C	...	...	...	...

3.6 Part List

No	Part	Materials	Color / Finish
1	Insulator	Thermal Plastic	UL94V-0, Black Color
2	Contact	Copper Alloy	Contact Area: Au0.05 μ m Min over Pd-Ni0.75 μ m Min over Ni1.0~5.0 μ m Lead Area: Au0.05 μ m Min over Ni1.0~5.0 μ m
3	Ground Plate	Stainless Steel	Nickel Plating
4	Mid Plate	Stainless Steel	Nickel Plating
5	Metal Shell	Stainless Steel	Nickel Plating

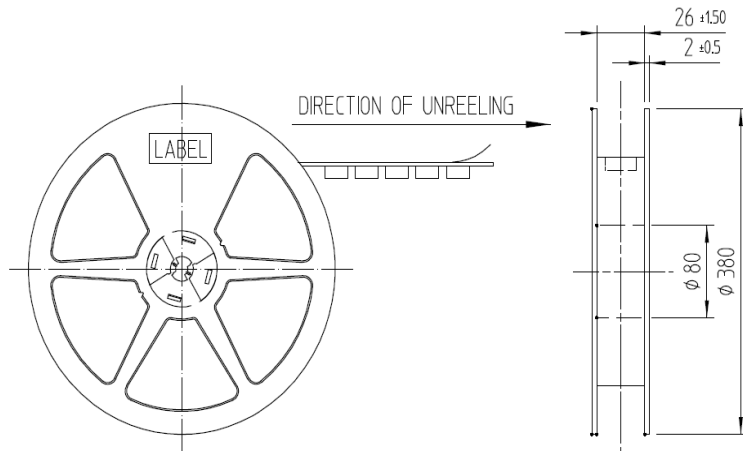


3.7 Configuration of Product Name

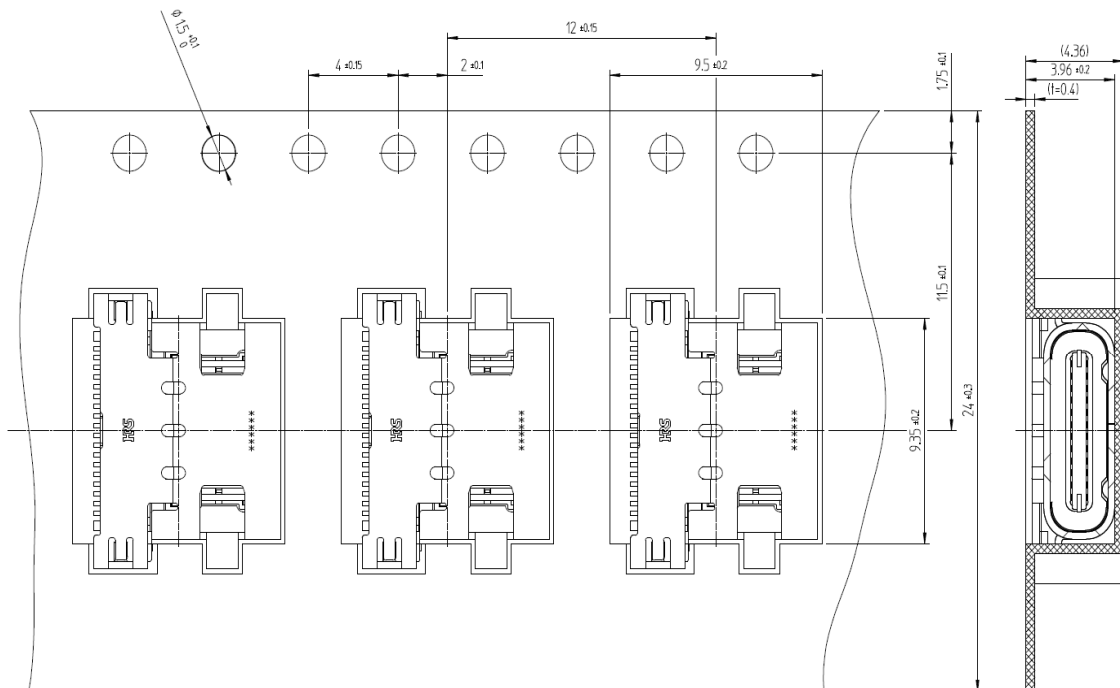
CX 70 M - 24 P 1
 (1) (2) (3) (4) (5) (6)

(1) Series Name		CX
(2) Soldering Type	60	Paddle Card
	70	Right angle Hybird (SMT+Dip)
	80	Straight SMT
	90	Right angle SMT
(3) Mounting Type	B	Top-mount
	M	Mid-mount
(4) Contact No		24
(5) Contact Type	P	Male contacts
	S	Female contacts
(6) Serial No		None or 1, 2, 3, ...

3.8 Reel Dimensions



3.9 Emboss Carrier Tape Dimensions



※ Emboss Tape was designed in accordance with EIA-481

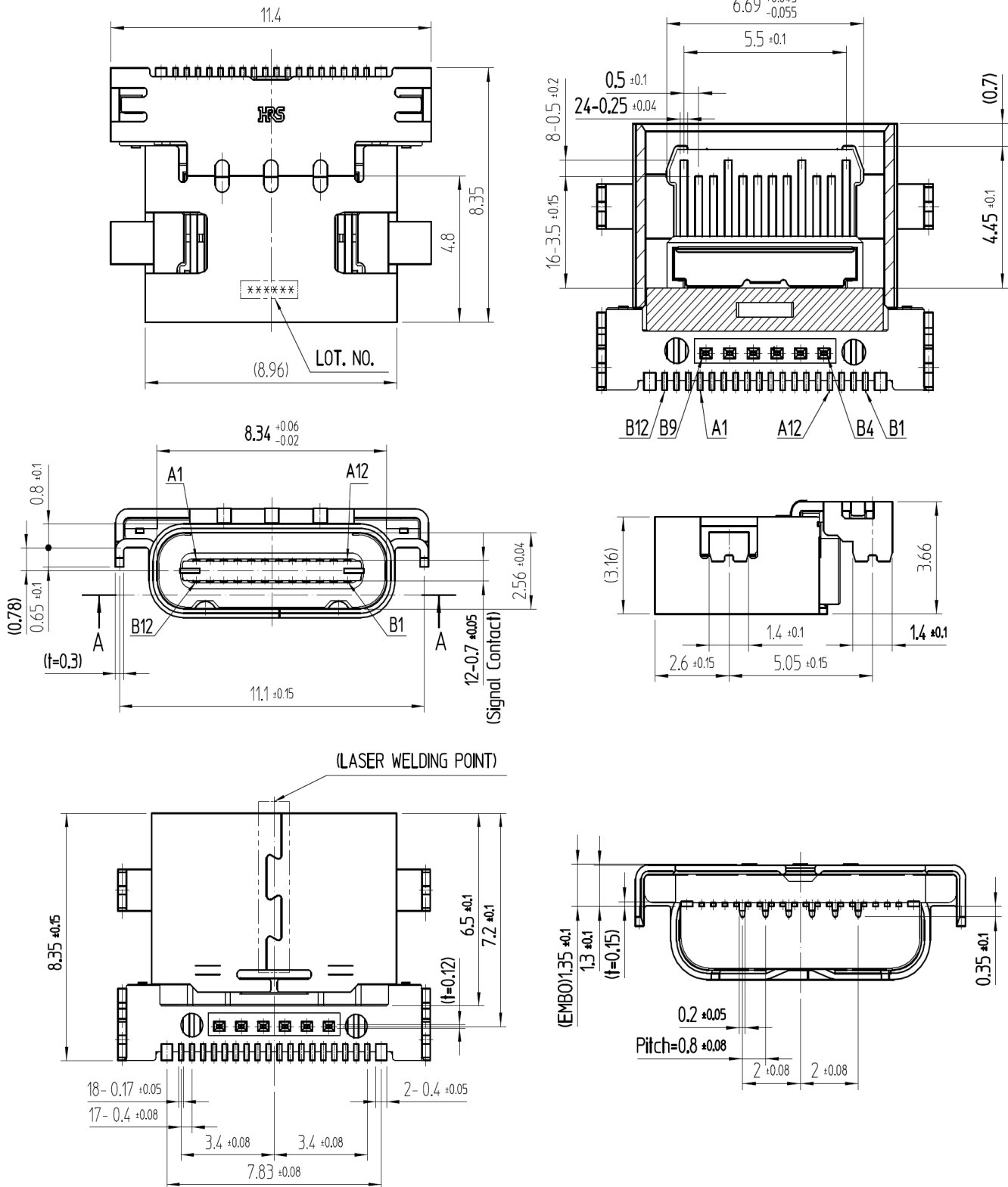
3.10 Packing Quantity Per Reel: 1,500 PCS

3.11 Peeling Strength:

Cover tape shall have a total peel strength of from 0.1N to 1.3N

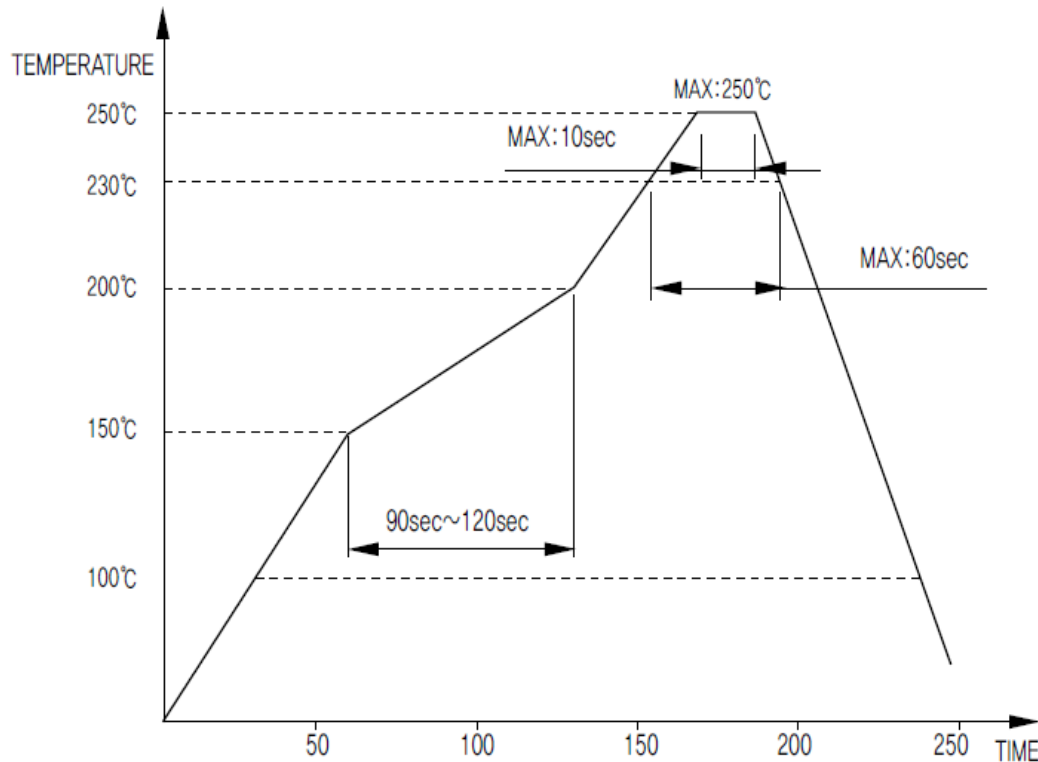
3.12 Product Dimensions

A-A



4. Notice for Soldering

4.1 Recommended Reflow Temperature Profile



- Profile measuring point

The temperature profile indicates the board surface temperature at the point of contacts with the connector terminals.

- Reflow cycles

Up to 2 cycles of reflow soldering are possible under the same conditions.

* Temperature between 1st and 2nd reflow must be cooled down to room temperature

- Reflow heating method and condition

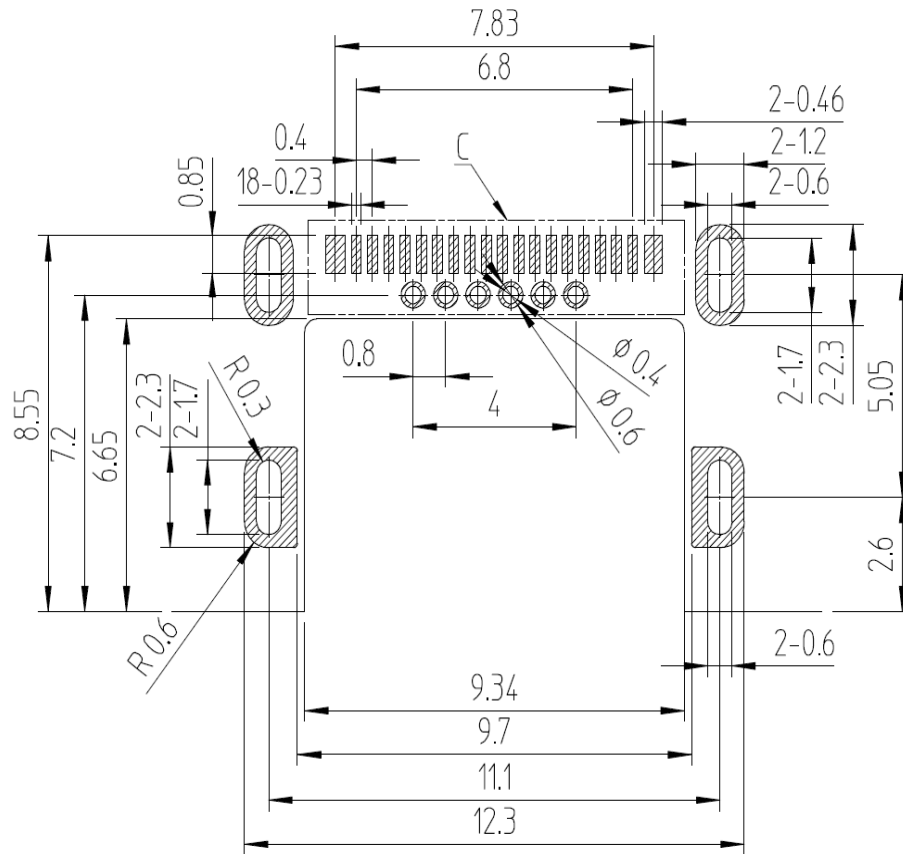
Far-infrared heater and hot convective blowers used in combination, normal atmosphere, or nitrogen atmosphere.

- Top surface of the contact leads may not covered by solder depending on reflow condition

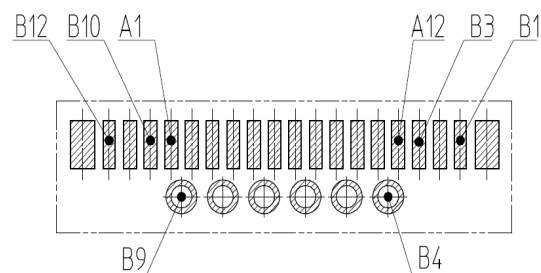
4.2 PCB Designing

RECOMMENDED PCB LAYOUT (TOP-VIEW)

(REFERENCE ONLY)



Detail C



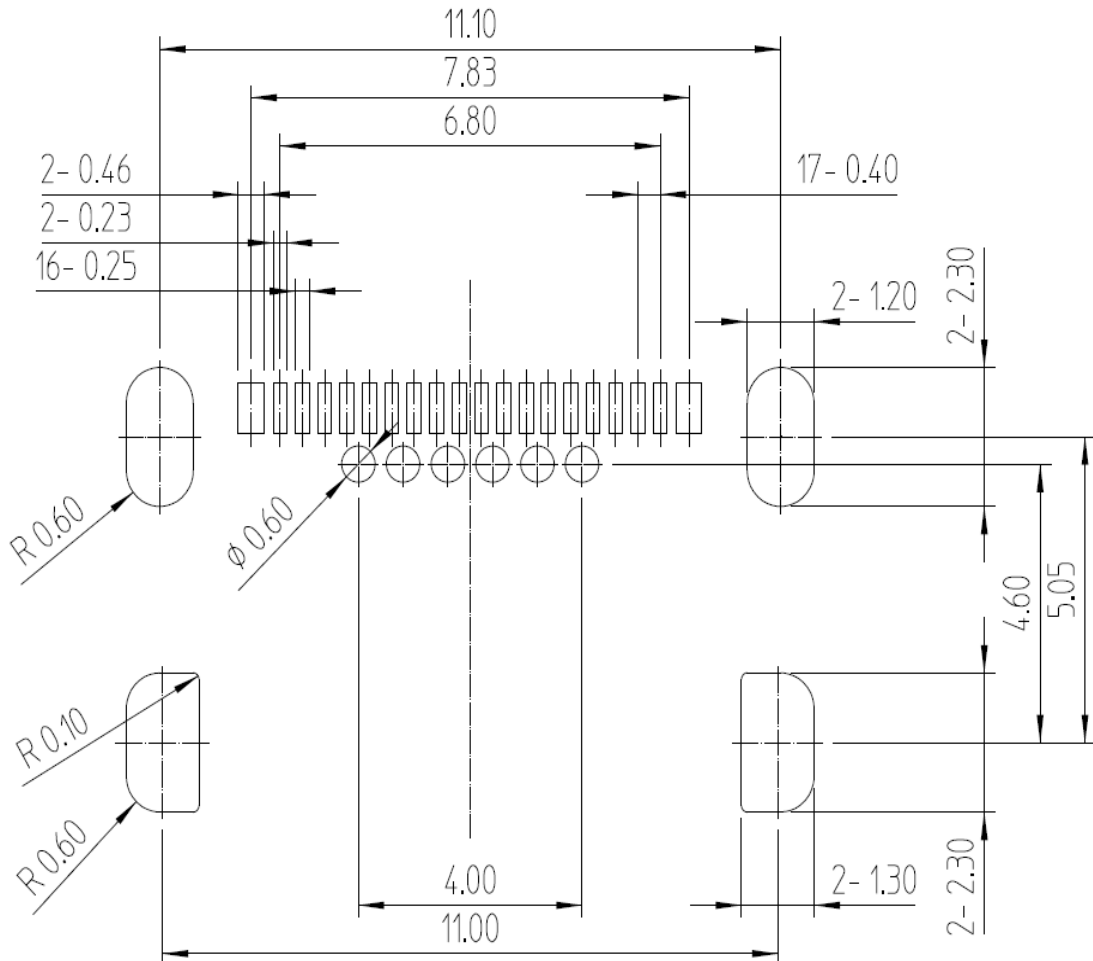
4.3 PCB Thickness

Recommended PCB thickness is 0.8mm.

4.4 Metal Mask Designing

RECOMMENDED METAL MASK LAYOUT (TOP-VIEW)

(REFERENCE ONLY)



- Metal mask thickness: 0.10mm
 - Open rate: 100%
- $$\text{Open rate(\%)} = \frac{\text{Opening size of mask}}{\text{Land size of PCB}}$$

4.5 Solder Paste

Lead-free solder paste

5 Notice for Connector Handling

5.1 Mating Plug

Please use USB Type-C Plug only

